Supplementary Information

Qualification of Pathology Foundation-Model Signals Across Cohorts, Sites and Clinical Endpoints

Jin Hyun Kim

Department of AI and Information Engineering, Gyeongsang National University, Jinju, South Korea

Draft submission metadata: the final author list, affiliations and corresponding-author details require author confirmation before submission.

Supplementary Methods and Results

This document reports the saved extended analyses that support, but do not enlarge, the claims in the main manuscript. Patient-level uncertainty and repeated tile-sampling variation are kept distinct. Missing or unevaluable outcomes were never treated as negative labels, and all endpoint names retain their source definitions.

Abbreviations. CONCH denotes CONtrastive learning from Captions for Histopathology; NADT denotes neoadjuvant androgen-deprivation therapy; PANDA denotes Prostate cANcer graDe Assessment; TCGA-PRAD denotes The Cancer Genome Atlas prostate adenocarcinoma cohort; and TCGA-CDR denotes the TCGA Pan-Cancer Clinical Data Resource. Area under the receiver operating characteristic curve (AUROC), time-dependent area under the curve (td-AUC), concordance index (C-index), integrated Brier score (IBS), progression-free interval (PFI), progression-free survival (PFS), disease-free survival (DFS), hematoxylin and eosin (H&E), copy-number alteration (CNA), confidence interval (CI), and 256-bit Secure Hash Algorithm (SHA256) are used below. Gene/protein symbols are expanded as follows: phosphatase and tensin homolog (PTEN), speckle type BTB/POZ protein (SPOP), androgen receptor (AR), tumor protein p53 (TP53), RB transcriptional corepressor 1 (RB1), serine peptidase inhibitor Kazal type 1 (SPINK1), ETS variant transcription factors 1 and 4 (ETV1 and ETV4), and ETS transcription factor ERG (ERG). The post-hoc Cox recurrence-risk score called “marker 7” in legacy artifacts is termed the post-hoc recurrence risk signal here. CH, EJ, G9, HC, KK, and YL are deidentified source institution codes rather than expanded institution names.

Supplementary Table S1 records the complete audited endpoint hierarchy without renaming reconstructed outcomes as official PFI.

Table S1: Endpoint hierarchy retained from the audited source. Rows E01–E02 are primary frozen-score qualification, E03 is secondary held-out clinical increment, and E04–E05 are exploratory recurrence analyses. The metric column states the estimand used for each endpoint class; hierarchy levels are not interchangeable.

ID	Hierarchy	Endpoint	Primary metric
E01	primary	continuous marker qualification	patient Spearman rho
E02	primary	binary marker qualification	patient AUROC
E03	secondary	PTEN/AR held-out increment	delta AUROC or delta R2
E04	exploratory	reconstructed with-tumor endpoint	C-index; td-AUC; IBS; calibration
E05	exploratory	strict recurrence-only endpoint	C-index; td-AUC
E06	exploratory	TCGA-CDR PFS	C-index; td-AUC; landmark AUC
E07	exploratory	TCGA-CDR DFS	C-index; td-AUC; landmark AUC

ID	Hierarchy	Endpoint	Primary metric
E08	exploratory	official TCGA-CDR PFI	C-index; endpoint concordance
E09	secondary	3-year and 5-year landmark	landmark C-index/AUROC

The hierarchy distinguishes what each analysis can answer. E01–E02 qualify frozen continuous and binary marker scores, E03 asks whether PTEN or AR adds held-out information beyond grade, and E04–E05 are exploratory recurrence analyses with different event definitions. Consequently, a result in an exploratory recurrence row cannot be used to upgrade a primary marker claim, and agreement between two endpoint fields does not erase their declared provenance.

The complete 17-test family

This subsection enumerates the complete multiplicity family so that the main-text findings can be interpreted against all saved discovery-family tests rather than a selected subset.

The multiplicity family contained 13 marker hypotheses and four nested/refit confounder audits. Continuous marker hypotheses used two-sided Spearman tests, binary hypotheses used two-sided Mann–Whitney tests, and the refit-permutation audit used the saved one-sided test for positive improvement. Benjamini–Hochberg false-discovery-rate adjustment was applied jointly to these 17 rows. External or cross-encoder evaluations of an already counted hypothesis were qualification checks rather than additional discovery-family rows.

Table S2: Complete saved 17-row revision-analysis multiplicity family. Every discovery-family row is shown rather than a selected subset. Effects and probability values are copied from the audited source; q is the Benjamini–Hochberg false-discovery-rate-adjusted value across all 17 rows.

Test	Metric	Effect	p	q
① NADT H&E -> Gleason	Spearman rho	0.478	0.00207	0.00703
② NADT H&E -> Phenotype	Spearman rho	0.8	9.63×10^{-10}	1.64×10^{-8}
③ NADT ERG -> Gleason	Spearman rho	0.524	0.00061	0.00346
④ TCGA-PRAD H&E -> PTEN loss	AUROC	0.632	0.000562	0.00346
⑤ TCGA-PRAD H&E -> SPOP mutation	AUROC	0.519	0.742	0.788
⑥ TCGA-PRAD H&E -> AR score	Spearman rho	0.195	0.00122	0.0052
TCGA-PRAD H&E -> TP53 mutation	AUROC	0.485	0.835	0.835
TCGA-PRAD H&E -> TP53 CNA loss	AUROC	0.562	0.0981	0.208
TCGA-PRAD H&E -> RB1 CNA loss	AUROC	0.536	0.363	0.441
TCGA-PRAD H&E -> SPINK1 high	AUROC	0.618	0.152	0.288
TCGA-PRAD H&E -> ETV1 altered	AUROC	0.429	0.245	0.347
TCGA-PRAD H&E -> ETV4 altered	AUROC	0.687	0.023	0.0559
TCGA-PRAD H&E -> ERG fusion status	AUROC	0.526	0.458	0.519
PTEN / grade-only	AUROC change	0.0195	0.203	0.314
AR / grade-only	R-squared change	0.0044	0.176	0.299
Post-hoc recurrence risk / grade-only	C-index change	0.0619	0.01	0.0283
Post-hoc recurrence risk / fully adjusted	C-index change	0.00163	0.305	0.399

Supplementary Table S2 reports every row in the saved 17-row revision-analysis multiplicity family; no selected subset is presented as the family.

The family records the tests as originally defined. Later endpoint-matched, same-patient and same-bootstrap-draw comparisons are reported below and govern the narrower recurrence interpretation; they do not retrospectively replace rows in the saved multiplicity family.

Detailed discrimination summaries

We first expand the cross-cohort grade and phenotype evidence underlying the transportable state assigned in the main Results.

Supplementary Figure S1 shows the saved aggregate discrimination and association estimates for grade and phenotype transfer across NADT, PANDA and PRECISE. No receiver operating characteristic coordinates were stored for these rows, so curves were not reconstructed from aggregate values. Missing intervals remain visibly unavailable.

The central pattern is directional continuity rather than uniformly precise effect-size estimation: the NADT-Prostate grade and phenotype signals retain their direction in the external PANDA subsets and in the PRECISE grade evaluation, but the PANDA and PRECISE rows without stored intervals do not establish the precision of those effects. Figure S1 therefore supports cross-resource transport of direction while leaving some target-cohort uncertainty unresolved.

Recurrence endpoints, paired bootstrap and common-source sensitivity

This subsection tests how the recurrence interpretation changes with endpoint definition, paired uncertainty estimation, and restriction to a common patient source frame.

The recurrence endpoints were analysed separately. On the 270-patient target frame, official TCGA-CDR PFI had 42 events and the reconstructed with-tumor endpoint had 57. Their event agreement was 0.870, Cohen's $\kappa = 0.569$, and follow-up-time Spearman $\rho = 0.849$. Official PFI and the cBioPortal TCGA-CDR PFS fields agreed exactly on the 270 common evaluable patients; this identity does not make the separately reconstructed endpoint equivalent to PFI. Among 192 patients shared with DFS, event agreement was 0.979 and $\kappa = 0.835$. Among 220 patients shared with the strict recurrence-only sensitivity, event agreement was 0.955 and $\kappa = 0.564$. These endpoint concordance results come from `models/pfi_endpoint_concordance.csv`. Cohen's kappa measures agreement after accounting for agreement expected by chance.

The paired survival analysis used 153 complete cases for each endpoint, with 30 reconstructed-endpoint events and 15 official-PFI events. Every reported model and contrast requested 2,000 patient-bootstrap resamples. The saved accounting was 2,000 valid and zero undefined bootstrap replicates (undefined fraction 0) for all 44 model-metric summaries and all 20 paired contrast-metric summaries. Zero undefined replicates is reported rather than silently assumed; another dataset could yield undefined concordance or Brier-score draws.

The post-hoc recurrence-risk source-retention audit compared all 60 seed cells on the saved source sets with the same cells restricted to the common-498 patient intersection. No cell crossed the C-index null, and all 65 paired contrast directions were preserved. The largest absolute cell shift was 0.023. This common-498 sensitivity limits source-retention confounding in the saved grid but does not provide an independent cohort or a causal representation comparison.

Supplementary Figure S2 displays the saved time-dependent AUROC and calibration summaries for the reconstructed with-tumor endpoint only. This legacy figure source is not Official TCGA-CDR PFI. Its endpoint identity and analysis frame were accepted only after the calibration quintiles at both horizons reconciled to the saved frozen-risk metadata row (270 patients and 57 events). The exploratory marker is not interpreted as an independently validated prognostic biomarker.

Figure S2 shows that the saved post-hoc score contains time-dependent discrimination and risk-stratification structure for the reconstructed endpoint. Because both panels use the same legacy 270-patient frame and reconstructed 57-event definition, they are complementary descriptions of one exploratory analysis, not two replications and not evidence of Official PFI calibration.

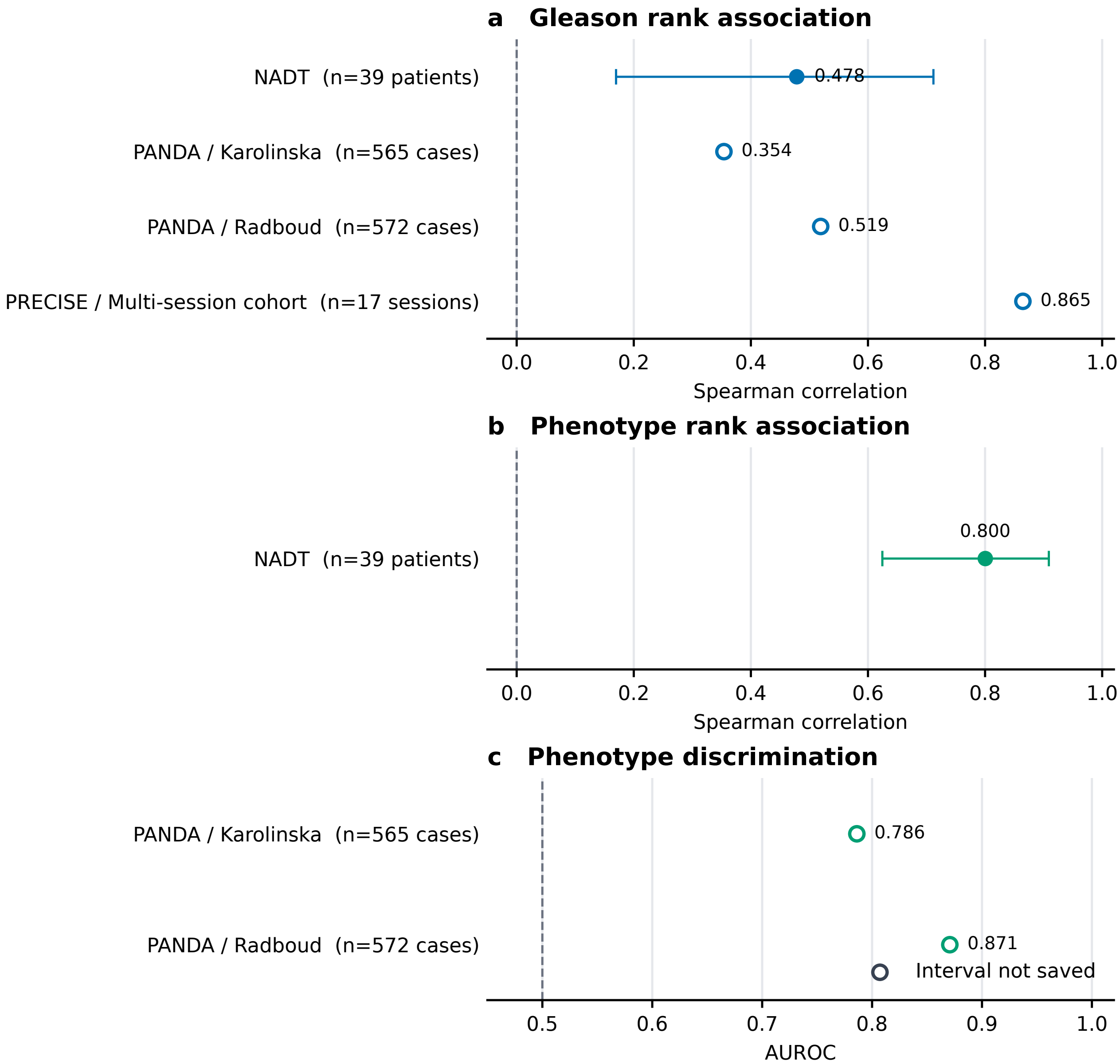


Figure S1: **Saved grade and phenotype transport details.** Aggregate patient-, case-image-, or session-level estimates are separated by metric: **a**, Gleason Spearman correlations in NADT-Prostate, PANDA, and PRECISE; **b**, NADT-Prostate phenotype Spearman correlation; and **c**, phenotype AUROC after transfer without refitting to the Karolinska and Radboud PANDA subsets. Filled points with whiskers show the saved estimate and patient-bootstrap interval; open points denote estimates for which no interval was saved. Dashed lines mark 0 for correlation and 0.5 for AUROC. Denominators and analysis units are printed for every row. Unavailable intervals were not treated as zero, and aggregate statistics were not converted into reconstructed curves.

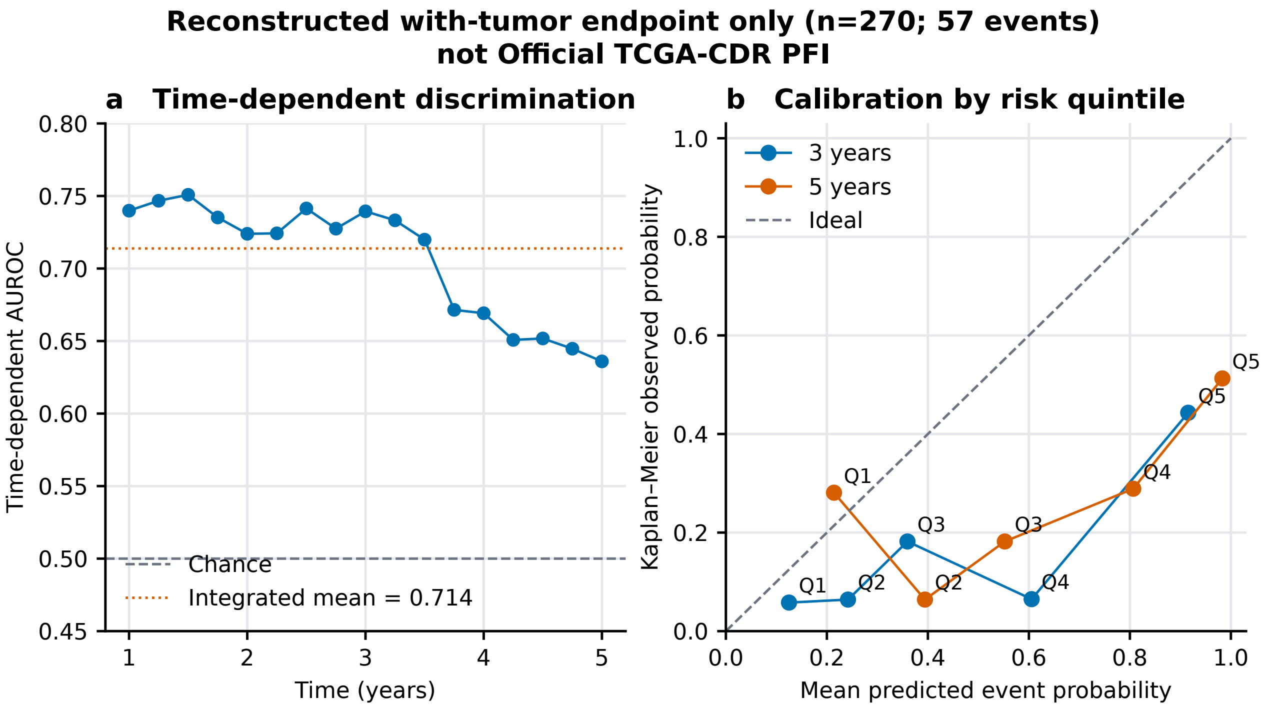


Figure S2: **Legacy reconstructed-endpoint post-hoc recurrence-risk summaries.** Values are rendered from saved legacy sources for the Reconstructed with-tumor endpoint (270 patients; 57 events), not Official TCGA-CDR PFI. **a**, Time-dependent AUROC; the dashed line is chance (0.5), and the dotted line is the saved integrated mean over 1–5 years. **b**, Mean predicted event probability versus Kaplan–Meier observed probability by risk quintile (Q1–Q5) at 3 and 5 years; the diagonal is ideal calibration and connecting lines are visual guides across quintiles. The endpoint identity, denominator, event count, and horizon-specific calibration totals were reconciled to the saved frozen-risk metadata. Kaplan–Meier is a nonparametric estimate of observed event probability; calibration compares predicted and observed probabilities; and risk quintiles divide patients into five ordered groups of predicted risk. This legacy summary is exploratory and supplies no Official-PFI calibration claim.

Full correlated stability analysis

We next expose the complete sensitivity grid behind the condensed stability decision in the main manuscript while preserving its correlated, non-confirmatory interpretation.

The saved grid has 72 configurations: six targets, two encoders, two physical scales per encoder and three tile budgets. Each configuration contains five repeated sampling seeds, giving 360 correlated seed cells. All settings reused the underlying cohorts and folds. The five-seed Student-t intervals therefore describe tile-sampling variation, not patient or population uncertainty. The saved paired-contrast table contains 390 seed-matched comparisons: 180 native-versus-shared-scale, 90 Virchow-versus-CONCH at shared scale and 120 64-versus-16-tile comparisons.

Supplementary Table S3 separates the observed seed range from the descriptive Student-t interval. Supplementary Figure S3 contains the complete configuration means followed by the paired contrasts. A setting-specific direction does not identify a causal effect of encoder, scale or tile count, and the 360 cells are not independent validations.

Table S3: Global seed-cell ranges and configuration-level null straddling recomputed from all 72 saved configurations. The last two columns count 12 configurations per target and distinguish the observed five-seed range from the descriptive Student-t interval.

Target	Global minimum	Global maximum	Range straddles	t-interval contains null
Gleason	0.271	0.646	0/12	0/12
Phenotype	0.845	0.965	0/12	0/12
PTEN	0.519	0.717	0/12	0/12
AR	0.136	0.297	0/12	0/12
SPOP	0.348	0.679	8/12	9/12
Recurrence risk	0.481	0.755	1/12	2/12

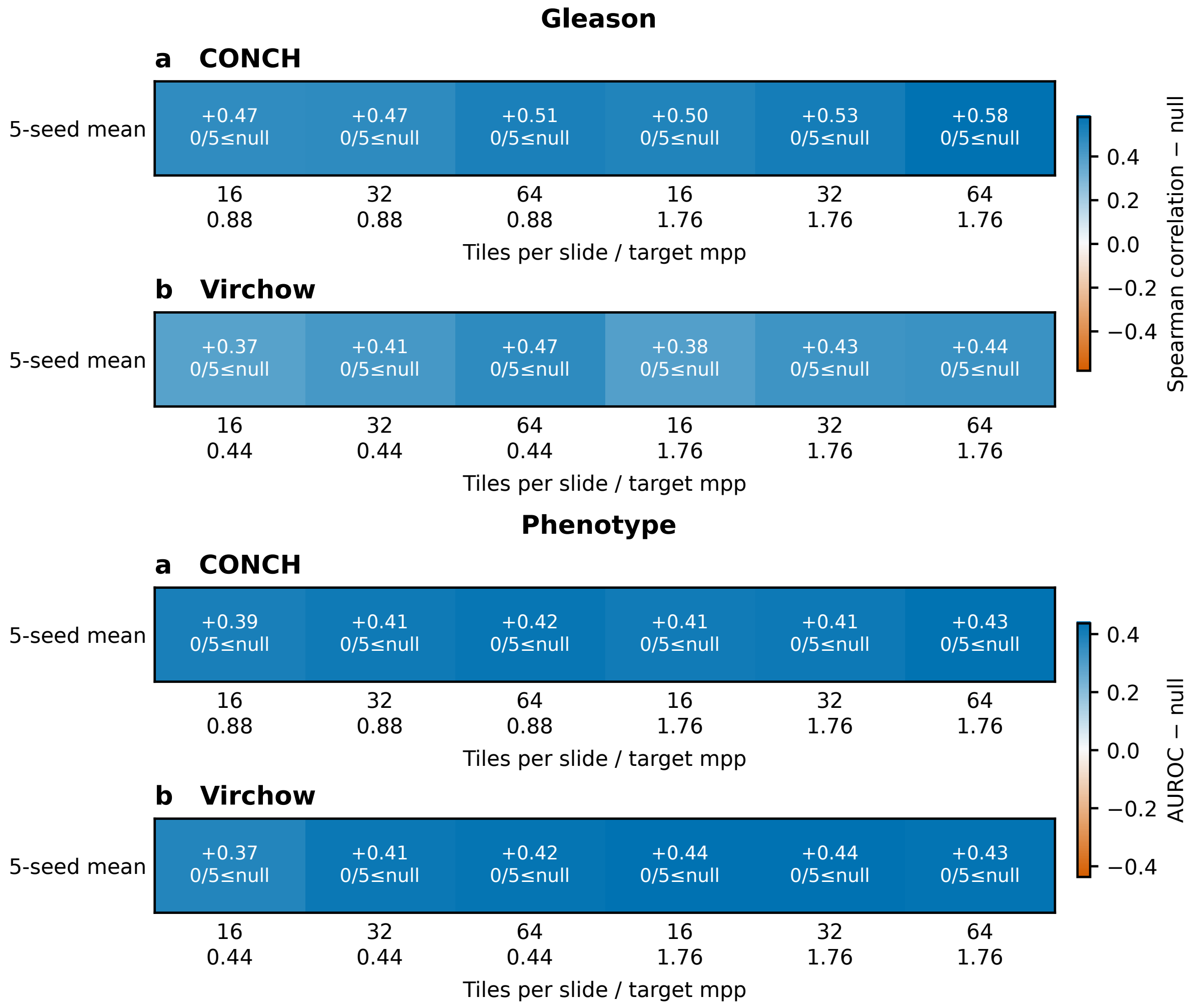
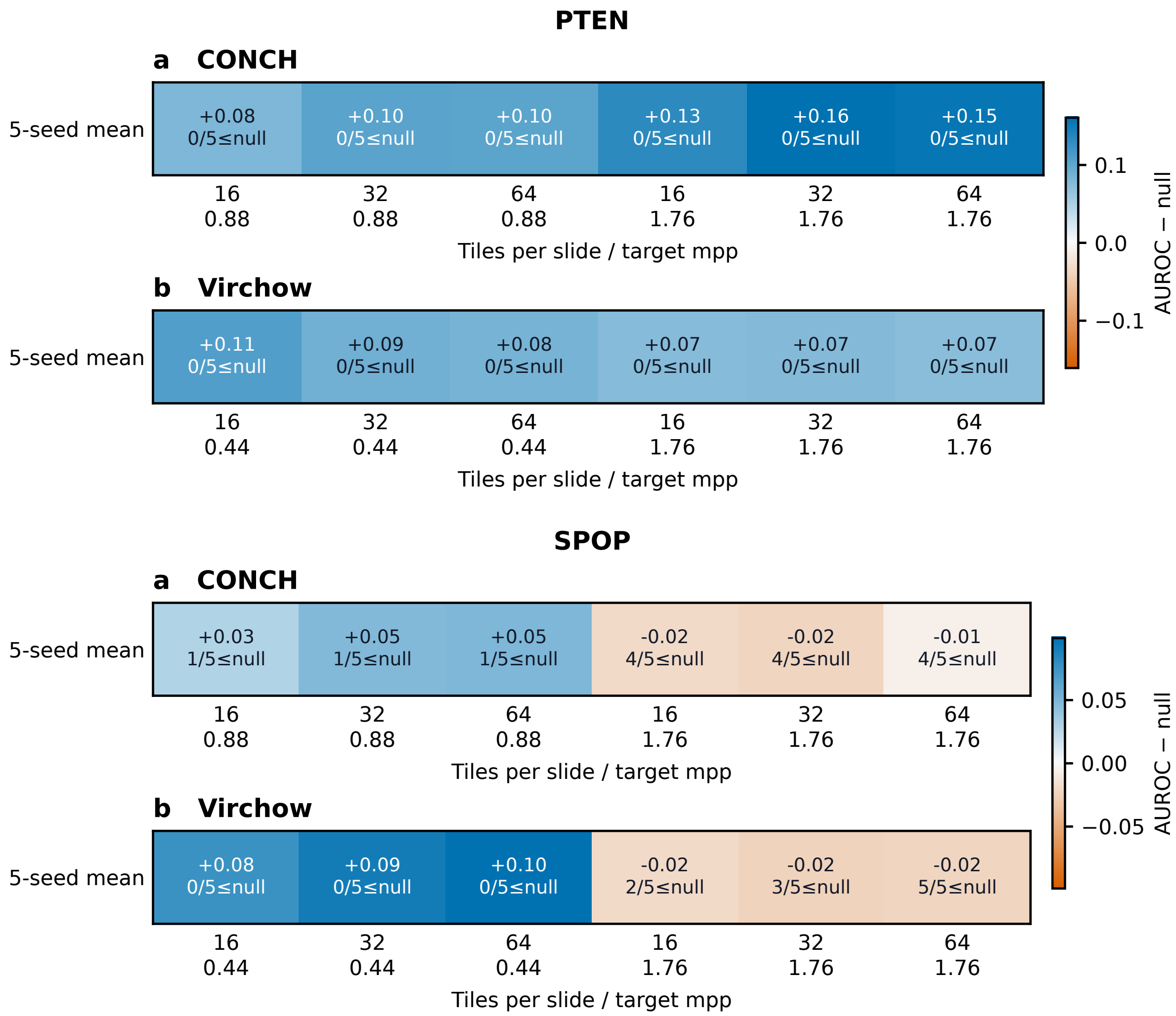
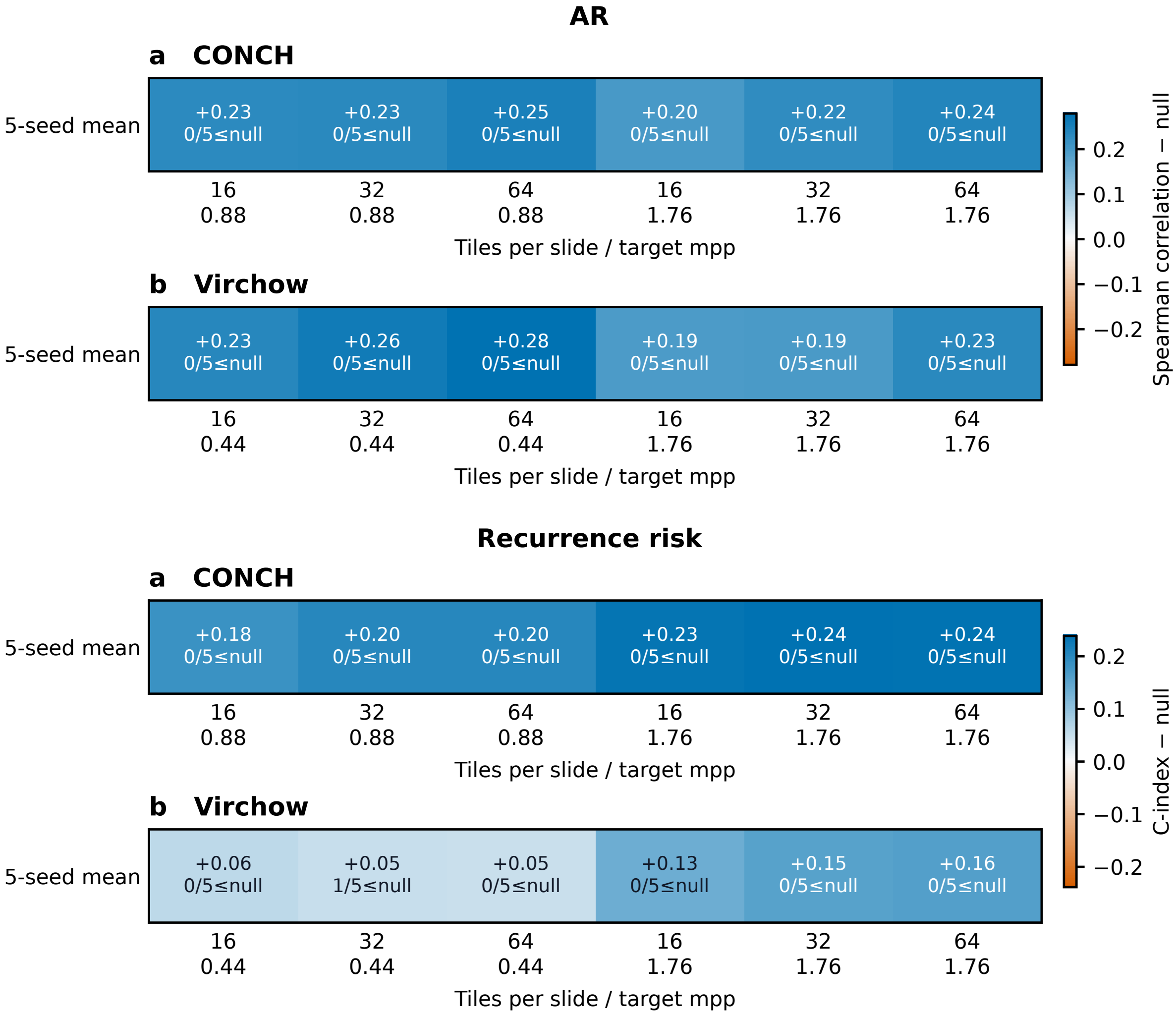


Figure S3: **Complete stability grids and paired contrasts.** Internal pages 1–6 display all 72 validated configuration means from the saved grid, two targets per physical supplementary page. For each target, panels **a–b** are CONCH and Virchow; columns combine 16, 32, or 64 tiles with the encoder-native physical scale or the shared 1.76 μm-per-pixel scale. Each cell reports the five-seed mean minus the marker-specific null and the number of seeds at or below that null; color is centred at zero separately for each target. Internal pages 7–9 display all 390 seed-matched contrasts: shared 1.76 μm-per-pixel minus native scale (180), Virchow minus CONCH at 1.76 μm per pixel (90), and 64 minus 16 tiles (120). The abscissa is metric $B - A$ and the dashed line is zero; orange points cross the marker-specific null whereas blue points remain on the same side. These are descriptive setting checks on reused cohorts and folds, not independent validations or population-confidence intervals.

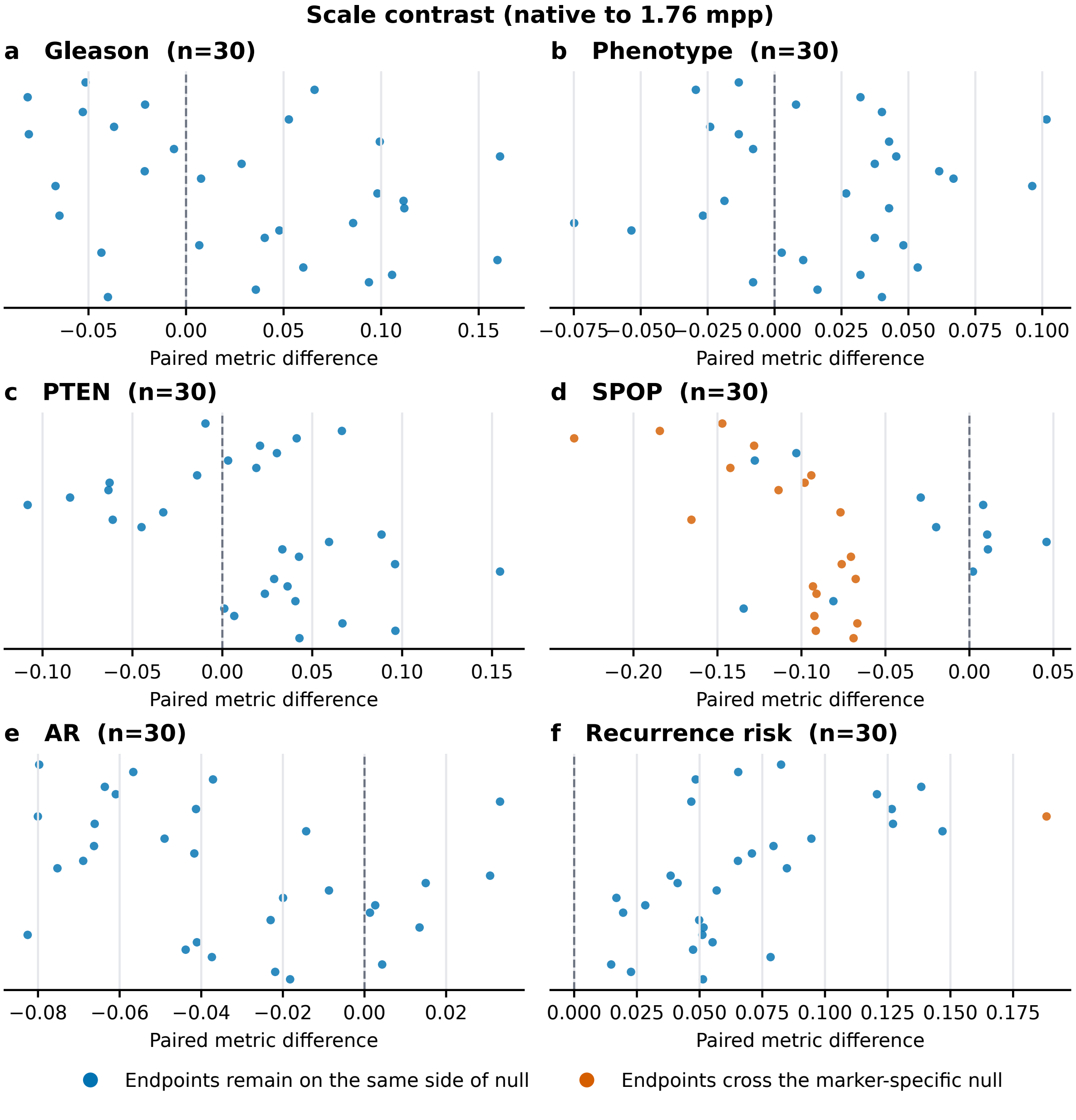
Supplementary Figure S3 (continued; heatmaps 3–4 of 6)



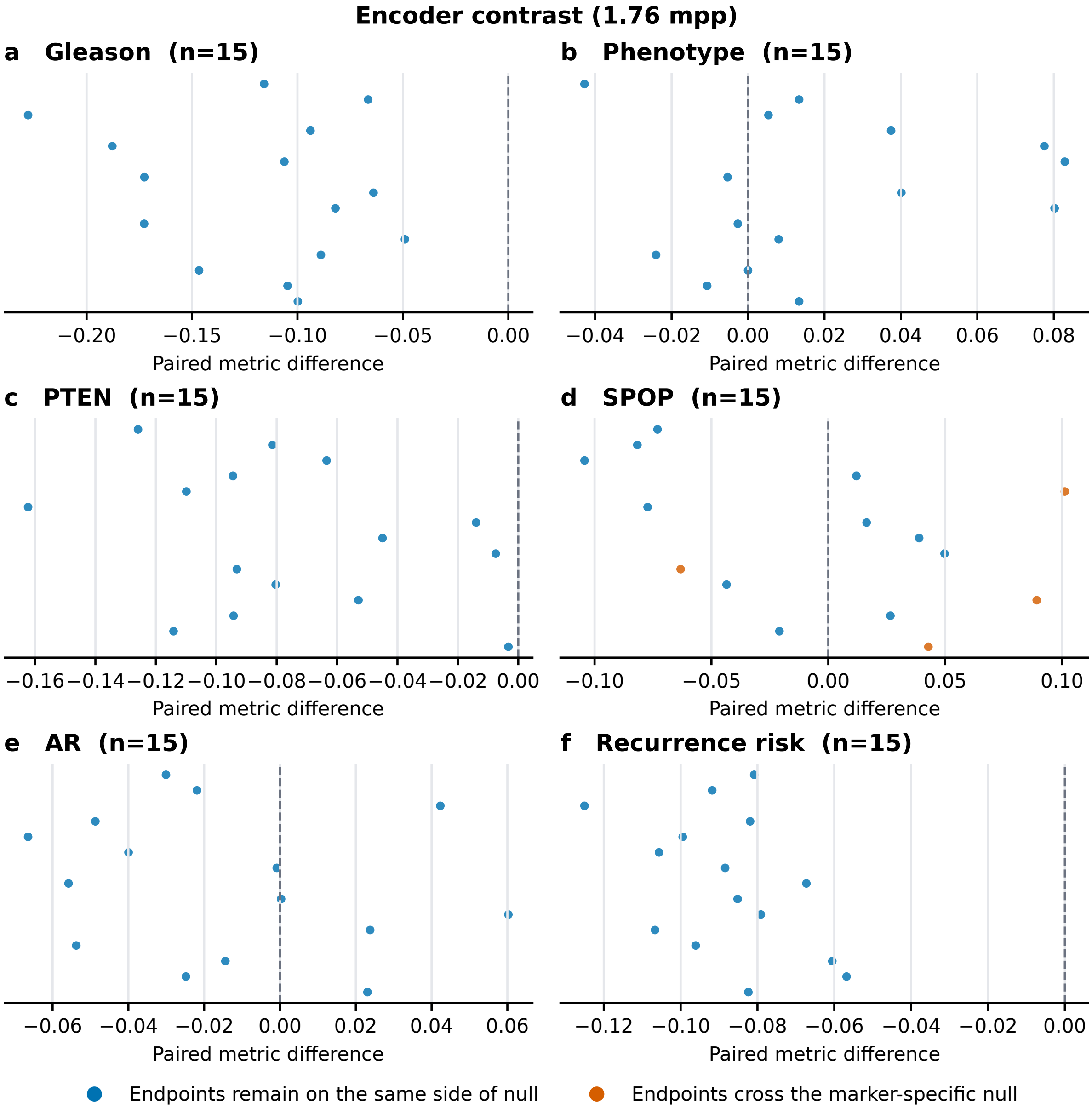
Supplementary Figure S3 (continued; heatmaps 5–6 of 6)



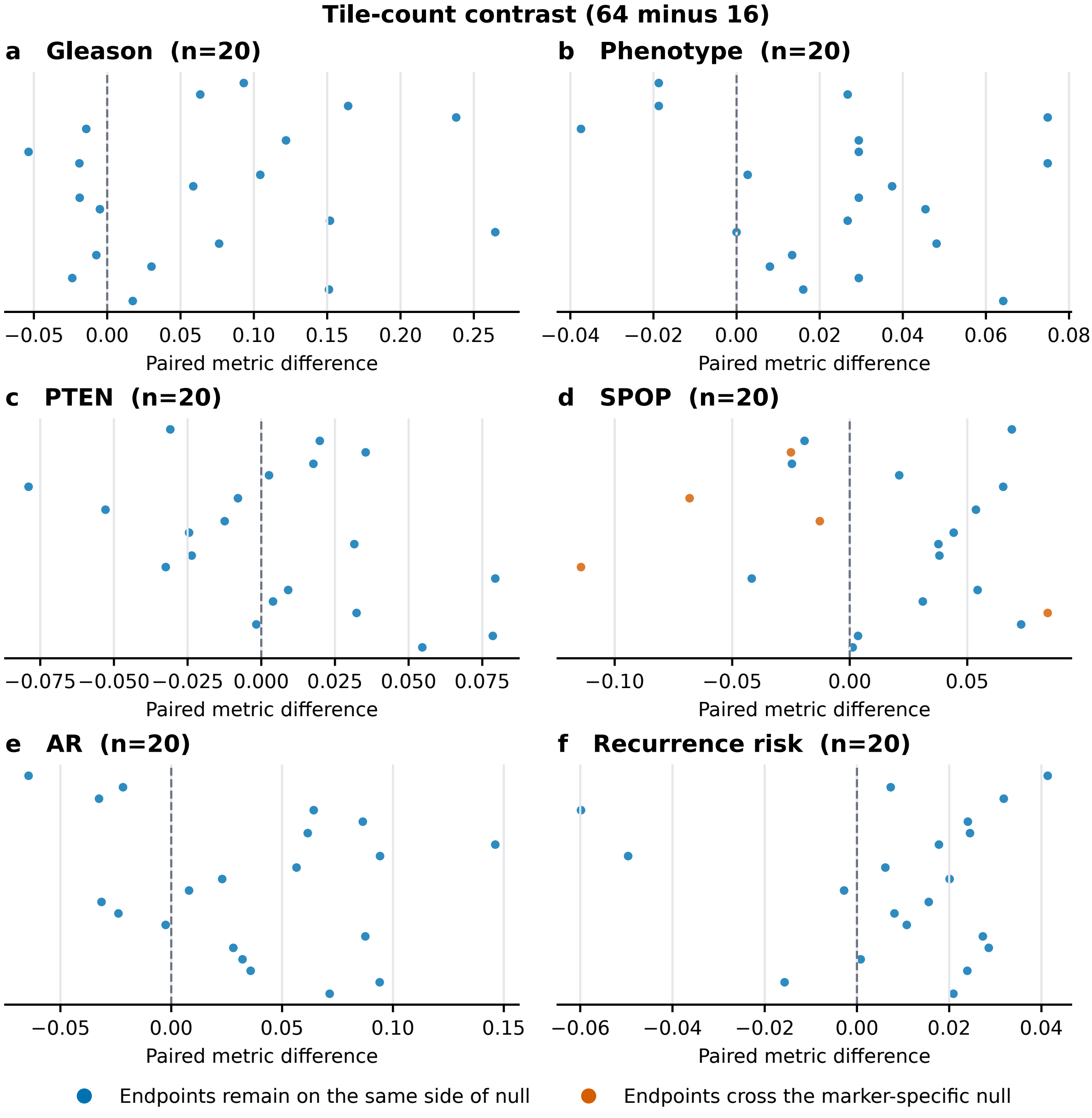
Supplementary Figure S3 (continued; scale contrasts)



Supplementary Figure S3 (continued; encoder contrasts)



Supplementary Figure S3 (continued; tile-budget contrasts)



Target-by-axis evidence coverage

This subsection makes the synthesis explicit by showing which evidence axis supports, restricts, or remains unavailable for each target.

The qualification framework deliberately preserves both positive evidence and evidentiary gaps. Supplementary Figure S4 shows the complete matrix of frozen-primary support, cross-cohort transport, clinical increment, site behavior, setting stability and endpoint fidelity for each target group. A supported cell refers only to the scope stated for that axis; it does not fill an unevaluated cell elsewhere. Supplementary Table S4 links every cell to the audited claim source, states the current interpretation and identifies the evidence required to change that state.

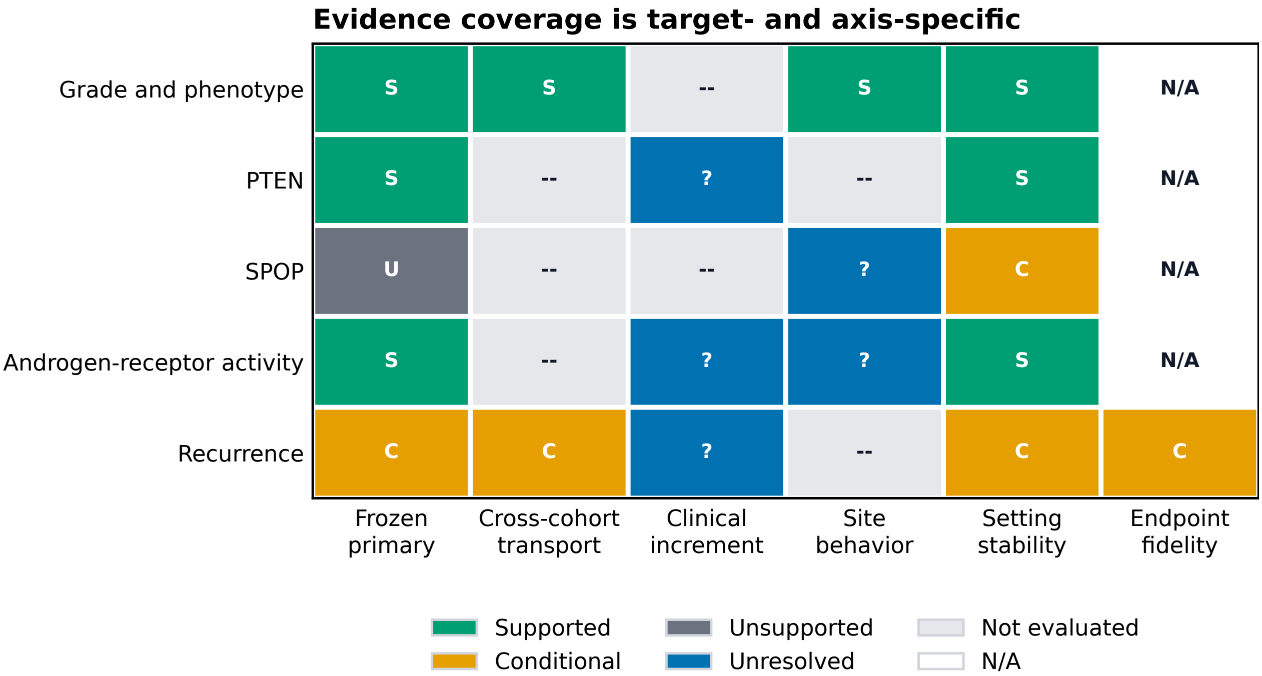


Figure S4: **Target-by-axis evidence coverage.** Each cell records the evidence state supported by the frozen analyses for one target row and one evidentiary-axis column. Columns separate frozen-primary support, cross-cohort transport, clinical increment, site behavior, setting stability, and endpoint fidelity. Cell labels are S (supported), C (conditional), U (unsupported in the frozen primary design), ? (evaluated but unresolved), – (not evaluated), and N/A (not applicable). A supported or conditional cell is bounded to its named axis and cannot fill another cell; not-evaluated and not-applicable cells are not negative evidence. The source-linked audit table below provides the claim basis and evidence needed to change every cell.

Read together, Figure S4 and Table S4 show that evidence completeness is target-specific. Grade and phenotype have the broadest support, including cross-cohort direction; PTEN and AR have supported within-cohort axes but unresolved or unevaluated clinical and site axes; SPOP remains unsupported or context-sensitive in the frozen design; and recurrence remains conditional on representation and endpoint. The matrix is therefore a map of qualified claims and open gaps, not a scorecard in which supported cells can compensate for missing ones.

Table S4: Target-by-axis qualification audit corresponding to Supplementary Figure S4. Each row links a target-specific evidence state to the audited claim source, states the current evidence, and specifies what evidence would be required to change that state. Not evaluated and not applicable are not negative results.

Target	Evidence axis	State	Claim	Current evidence	Next evidence needed
Grade and phenotype	Frozen primary	supported	C02	Source-cohort grade and phenotype signals were supported.	Retain the locked source probes and analysis units.
Grade and phenotype	Cross-cohort transport	supported	C02	Direction was retained without refitting in PANDA and was concordant with PRECISE grade evaluation.	Add patient-level uncertainty where source data permit and test further institutions.
Grade and phenotype	Clinical increment	not evaluated	C02	No incremental clinical-covariate claim was defined for these morphology-linked targets.	Define a clinically meaningful reference model before making an increment claim.
Grade and phenotype	Site behavior	supported	C02	Transfer directions were retained at both PANDA institutions; this was not a causal site analysis.	Use site-specific patient-level intervals and acquisition metadata in additional cohorts.
Grade and phenotype	Setting stability	supported	C02;C08	Correlated setting audits retained the transported directions.	Confirm stability on independently sampled patients rather than reused settings.
Grade and phenotype	Endpoint fidelity	not applicable	C02	No time-to-event endpoint was used for these targets.	Not applicable to the present grade and phenotype claims.
PTEN	Frozen primary	supported	C03	The within-TCGA frozen-primary association was above chance.	Repeat the locked evaluation in an external cohort with compatible PTEN labels.
PTEN	Cross-cohort transport	not evaluated	C03	No compatible external PTEN cohort was available in this study.	Acquire an external cohort with target-matched PTEN assays.
PTEN	Clinical increment	unresolved	C03	Held-out increment beyond grade was not established.	Test a locked image increment against grade and fuller clinical covariates in new patients.
PTEN	Site behavior	not evaluated	C03	The manuscript does not establish external site transport for PTEN.	Use a multisite external design with site and acquisition metadata.
PTEN	Setting stability	supported	C03;C08	The within-cohort direction remained above chance across the evaluated grid.	Confirm the direction under independent cohort resampling.
PTEN	Endpoint fidelity	not applicable	C03	PTEN loss was a binary molecular target rather than a clinical time-to-event endpoint.	Not applicable to the present PTEN claim.
SPOP	Frozen primary	unsupported in frozen design	C04	The frozen primary configuration did not support the proposed effect.	Increase target-appropriate sampling without changing the frozen primary interpretation.
SPOP	Cross-cohort transport	not evaluated	C04	No compatible external SPOP cohort was evaluated.	Test a locked probe in an external cohort with mutation labels and tumor-content control.
SPOP	Clinical increment	not evaluated	C04	A grade-independent SPOP increment was not established in the approved analysis set.	Evaluate incremental value with patient-disjoint nested analysis if scientifically justified.
SPOP	Site behavior	unresolved	C04	Defined site intervals included or touched chance and one site was single-class.	Collect larger site-balanced data with assay and acquisition metadata.
SPOP	Setting stability	context-sensitive	C04;C08	The result crossed the null across multiple correlated settings.	Resolve tumor sampling and representation sensitivity in independent data.

Target	Evidence axis	State	Claim	Current evidence	Next evidence needed
SPOP	Endpoint fidelity	not applicable	C04	SPOP mutation was a binary molecular target rather than a clinical time-to-event endpoint.	Not applicable to the present SPOP claim.
Androgen-receptor activity	Frozen primary	supported	C05	The pooled frozen-primary direction was positive.	Retain the pooled result as within-cohort evidence only.
Androgen-receptor activity	Cross-cohort transport	not evaluated	C05	No compatible external androgen-receptor activity cohort was evaluated.	Use an external cohort with a compatible activity assay.
Androgen-receptor activity	Clinical increment	unresolved	C05	Held-out grade-independent increment was not established.	Test a locked incremental model against grade and fuller clinical structure.
Androgen-receptor activity	Site behavior	unresolved	C05	Every stored site-specific interval crossed zero and acquisition effects could not be separated.	Disentangle site from scanner stain preparation and case mix.
Androgen-receptor activity	Setting stability	supported	C05;C08	The positive direction was retained across the correlated setting grid.	Confirm with independently sampled sites and patients.
Androgen-receptor activity	Endpoint fidelity	not applicable	C05	Androgen-receptor activity was a continuous molecular target rather than a time-to-event endpoint.	Not applicable to the present activity-score claim.
Recurrence	Frozen primary	context-sensitive	C06	The transferred post-hoc risk signal depended on representation and endpoint.	Lock the source model before a new independent target-cohort evaluation.
Recurrence	Cross-cohort transport	context-sensitive	C06	LEOPARD-to-TCGA transfer was present in some settings but was not endpoint-invariant.	Test the locked score in an external cohort with a harmonized endpoint.
Recurrence	Clinical increment	unresolved	C07	Full clinical and site comparisons did not establish robust incremental value.	Use an adequately powered same-patient comparison against a clinical reference locked before evaluation.
Recurrence	Site behavior	not evaluated	C06;C07	Site was included in fuller models but site-specific recurrence transport was not isolated.	Evaluate site transport with sufficient events per site.
Recurrence	Setting stability	context-sensitive	C06;C08	Effect size and direction depended on encoder scale and related settings.	Use nested locked model selection and independent validation.
Recurrence	Endpoint fidelity	context-sensitive	C06;C07	Reconstructed recurrence and official PFI produced different evidentiary conclusions.	Harmonize endpoint definitions and retain endpoint-specific reporting.

Supplementary evidence atlas

This subsection consolidates the breadth of the saved evaluation program in forms that are complementary to the point-and-interval plots in the main manuscript. It does not add new cohorts, refit a frozen representation, or treat repeated settings as independent experiments. Supplementary Table S5 identifies the analysis unit, denominator, repeated structure, and scientific role of every major evidence component. Counts from different rows must not be added because patients, slides, folds, and settings can overlap.

Table S5: Analysis-frame inventory. Each row identifies the resource, target, analysis unit, saved denominator or repeated structure, and scientific role of one evidence component. Counts must not be summed as independent patients or validations because patients, slides, folds, and settings can overlap across rows.

Evidence component	Resource	Target	Analysis unit	Saved frame	Role
Source fitting	NADT-Prostate	Grade; phenotype	Patient	39	Source-cohort probe fitting and patient aggregation
External transfer	PANDA Karolinska	Grade; phenotype	Case image	565; 469 tumor for phenotype	Transfer without target-cohort refitting
External transfer	PANDA Radboud	Grade; phenotype	Case image	572; 483 tumor for phenotype	Transfer without target-cohort refitting
Additional grade evaluation	PRECISE	Grade	Imaging session	17 evaluable sessions	Independent resource and session-level analysis unit
Molecular primary	TCGA-PRAD	PTEN; SPOP; AR	Patient	273	Frozen pooled association
Clinical increment	TCGA-PRAD	PTEN; AR	Patient	273	Nested held-out increment beyond grade
Site audit	TCGA-PRAD	AR	Slide with patient clustering	300 slides; 273 patients; 6 sites	Site-specific uncertainty, not a causal scanner analysis
Recurrence transfer	TCGA-PRAD	Post-hoc recurrence risk	Patient	270; 57 reconstructed and 42 Official-PFI events	Endpoint-specific frozen-risk transfer
Paired recurrence increment	TCGA-PRAD	Post-hoc recurrence risk	Complete-case patient	153; 30 reconstructed and 15 Official-PFI events	Same-patient, same-bootstrap-draw model contrasts
Correlated setting audit	Multiple source cohorts	Six targets	Configuration and seed cell	72 configurations; 360 cells; 390 paired contrasts	Sensitivity analysis; cells are not independent validations

Table S5 demonstrates breadth across cohorts, analysis units, molecular and recurrence targets, and correlated setting checks. It does not imply that its rows are independent: the same patient, slide, fold, or saved representation can contribute to more than one row, so the listed counts must be interpreted within their row-specific analysis frames.

Supplementary Figure S5 exposes all saved paired setting differences rather than only their global ranges. Each point is one seed-matched comparison, the violin shows its descriptive distribution, and the embedded box shows the median and interquartile range. The three panels represent shared-minus-native physical scale, Virchow-minus-CONCH encoder, and 64-minus-16-tile contrasts. Orange points identify pairs whose endpoints lie on opposite sides of the marker-specific null; they do not represent independent hypothesis tests. The complete numerical summaries are reported in Supplementary Table S6.

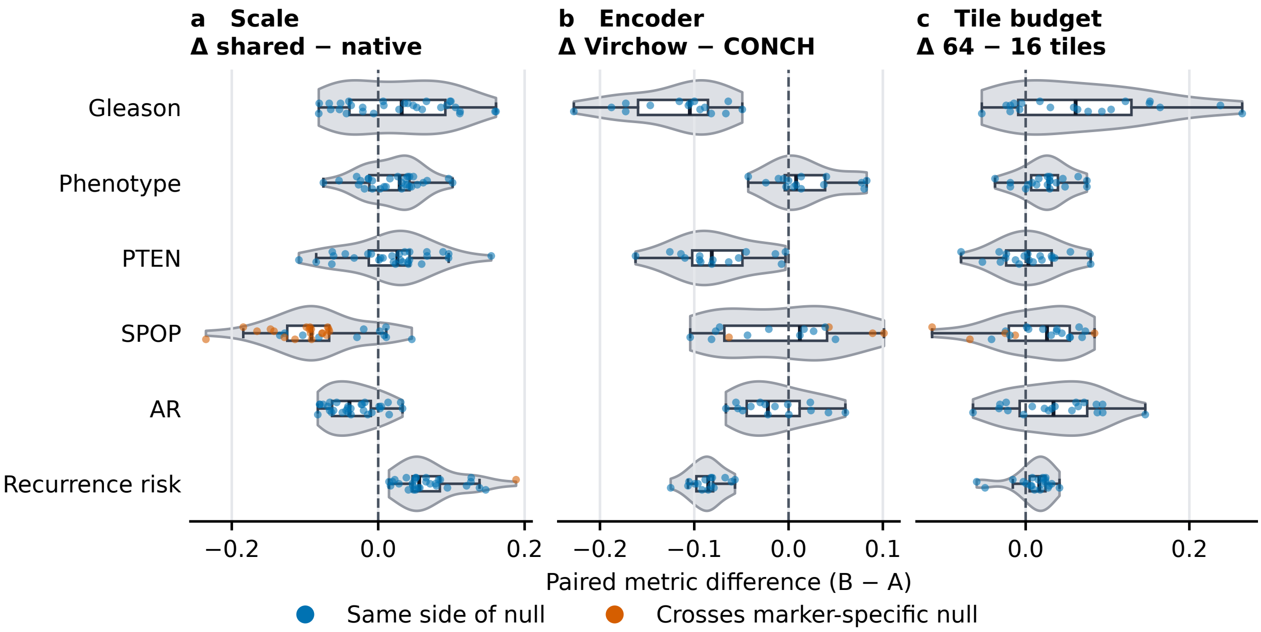


Figure S5: **Distributions of all seed-matched stability contrasts.** **a**, Shared 1.76-micrometre-per-pixel scale minus each encoder's native scale. **b**, Virchow minus CONCH at the shared scale. **c**, 64 minus 16 sampled tiles per slide. Horizontal violins show the distribution of saved paired metric differences; white boxes span the interquartile range, black lines mark medians, and individual dots are the complete seed-matched contrasts. Blue pairs remain on the same side of the relevant null and orange pairs cross it. Positive values favor the second-named setting under the displayed B-minus-A convention. All pairs reuse cohorts and folds and therefore describe setting sensitivity rather than independent validation evidence.

Table S6: Descriptive summary corresponding to Supplementary Figure S5. All 390 seed-matched setting contrasts are partitioned by comparison family and target. Differences are B minus A; medians, interquartile ranges, and null-crossing counts summarize correlated sensitivity comparisons rather than independent replicates or hypothesis tests.

Comparison	Target	Pairs	Median	Interquartile range	Null crossings
Shared 1.76 mpp minus native	Gleason	30	+0.032	-0.039 to +0.092	0/30
Shared 1.76 mpp minus native	Phenotype	30	+0.029	-0.012 to +0.043	0/30
Shared 1.76 mpp minus native	PTEN	30	+0.026	-0.013 to +0.043	0/30
Shared 1.76 mpp minus native	AR	30	-0.039	-0.063 to -0.010	0/30
Shared 1.76 mpp minus native	SPOP	30	-0.091	-0.124 to -0.067	19/30
Shared 1.76 mpp minus native	Recurrence risk	30	+0.056	+0.047 to +0.084	1/30

Comparison	Target	Pairs	Median	Interquartile range	Null crossings
Virchow minus CONCH at 1.76 mpp	Gleason	15	-0.105	-0.160 to -0.085	0/15
Virchow minus CONCH at 1.76 mpp	Phenotype	15	+0.008	-0.004 to +0.039	0/15
Virchow minus CONCH at 1.76 mpp	PTEN	15	-0.081	-0.102 to -0.049	0/15
Virchow minus CONCH at 1.76 mpp	AR	15	-0.022	-0.044 to +0.012	0/15
Virchow minus CONCH at 1.76 mpp	SPOP	15	+0.012	-0.068 to +0.041	4/15
Virchow minus CONCH at 1.76 mpp	Recurrence risk	15	-0.085	-0.098 to -0.080	0/15
64 minus 16 tiles	Gleason	20	+0.061	-0.009 to +0.129	0/20
64 minus 16 tiles	Phenotype	20	+0.028	+0.007 to +0.039	0/20
64 minus 16 tiles	PTEN	20	+0.003	-0.024 to +0.032	0/20
64 minus 16 tiles	AR	20	+0.034	-0.007 to +0.075	0/20
64 minus 16 tiles	SPOP	20	+0.026	-0.021 to +0.054	5/20
64 minus 16 tiles	Recurrence risk	20	+0.017	+0.005 to +0.024	0/20

Figure S5 and Table S6 make the heterogeneity behind the condensed stability decision visible. Grade, phenotype, PTEN, and AR have no null-crossing pairs in the saved contrast families, whereas SPOP contains crossings for scale, encoder, and tile-budget comparisons. The recurrence-risk signal has one scale crossing and a consistently negative Virchow-minus-CONCH contrast on this grid. These patterns identify settings that require qualification; they do not rank encoders causally or turn repeated seeds into independent experiments.

Supplementary Figure S6 separates three notions that can otherwise be hidden by a single endpoint label: event agreement, chance-corrected agreement, and follow-up-time rank correlation. The event-pair composition shows where apparent agreement arises from shared non-events and where the event indicators disagree. Exact agreement between Official TCGA-CDR PFI and the saved TCGA-CDR PFS fields reflects their source-field identity in this analysis; it is not an independent replication. Conversely, high raw agreement with other endpoints does not make their event definitions equivalent.

Table S7: Saved endpoint concordance corresponding to Supplementary Figure S6, with Official TCGA Pan-Cancer Clinical Data Resource progression-free interval (PFI) as the reference. Common N is the shared evaluable frame; event counts are Official PFI/comparison; agreement is raw event agreement, kappa is chance-corrected event agreement, and time rho is Spearman follow-up-time correlation.

Comparison endpoint	Common N	Events	Agreement	Kappa	Time rho
Reconstructed with-tumor	270	42/57	0.870	0.569	0.849
TCGA-CDR PFS	270	42/42	1.000	1.000	1.000
TCGA-CDR DFS	192	15/11	0.979	0.835	0.994
Strict recurrence-only	220	17/7	0.955	0.564	0.951

Figure S6 and Table S7 show why raw agreement alone is insufficient. The exact Official-PFI and TCGA-CDR-PFS row reflects identical saved source fields in this analysis, while the reconstructed and strict-recurrence rows combine high raw agreement with only moderate chance-corrected agreement. Recurrence conclusions are therefore reported by endpoint rather than pooled under a single label.

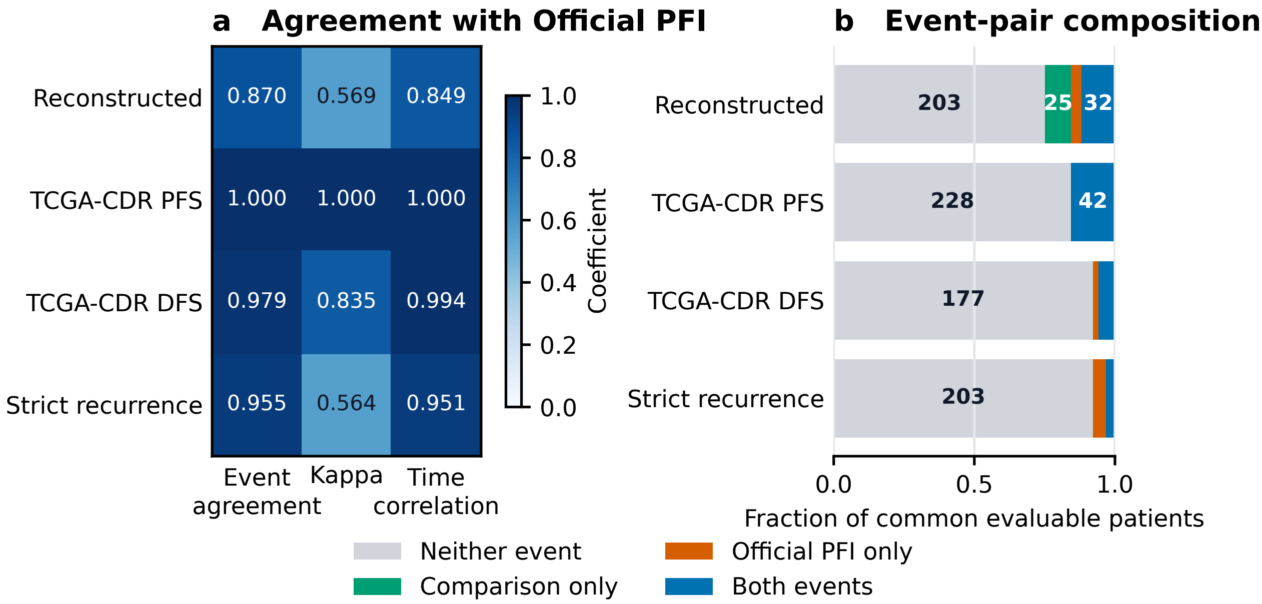


Figure S6: **Endpoint concordance relative to Official TCGA-CDR progression-free interval (PFI). a**, Heatmap of raw event agreement, Cohen's kappa, and Spearman correlation of follow-up times among common evaluable patients. Cohen's kappa adjusts event agreement for agreement expected by chance. **b**, Composition of the corresponding event pairs; numbers within sufficiently large segments are patient counts. Grey denotes neither endpoint recording an event, green the comparison endpoint only, orange Official PFI only, and blue both endpoints. Differences in evaluable frames and event definitions remain part of the interpretation.

AR site uncertainty and SPOP detectable-effect calculations

We then provide the detailed checks that bound the site-specific AR interpretation and the effect sizes that the frozen SPOP analysis was positioned to detect.

The AR site audit estimated slide-level Spearman correlations and used patient-cluster bootstrap intervals. Six leave-one-site-out rows were available: CH, −0.128 (−0.561 to 0.350); EJ, 0.127 (−0.121 to 0.363); G9, 0.195 (−0.258 to 0.598); HC, 0.217 (−0.072 to 0.471); KK, 0.265 (−0.087 to 0.558); and YL, 0.227 (−0.135 to 0.541). Every site-specific interval included zero. The positive pooled patient-level estimate therefore did not establish transport at any individual site, and the data did not identify scanner or stain as a cause.

For SPOP, the saved planning calculation used 29 positive and 244 negative patients, a two-sided $\alpha = 0.05$ test and a fixed-score Hanley–McNeil normal approximation. The minimum detectable AUROC was 0.661 at 80% power and 0.685 at 90% power. These values omit probe-training uncertainty, out-of-fold dependence, site heterogeneity, setting selection and multiplicity. They are planning approximations, not effect-exclusion bounds; they do not show that smaller effects or sampling dilution are absent. In this calculation, "minimum detectable" means the smallest assumed AUROC that reaches the specified power under the stated assumptions, not the smallest biologically possible effect.

Provenance and reproducibility scope

The final subsection identifies the saved records supporting computational traceability and distinguishes that traceability from external or clinical validation.

The extended results were read from saved CSV sources rather than recomputed from remembered values. Stability lineage is recorded in `models/stability_run_manifest.csv` and `models/stability_qc_report.json`; recurrence lineage is recorded in the paired survival and common-source run configurations and manifests; AR and SPOP source hashes are recorded in `models/ar_spop_evidence_manifest.csv`. The manuscript source tables and figure renderers are linked by the submission figure manifest. The PRECISE clinician review source was kept immutable and its SHA256 was checked against `c1dd522b4ff4f233b3a23630bf9074da881bb7b9145996fc47c3383a0448d2a3`. A SHA256 hash is a file fingerprint used to detect byte-level changes, while a manifest is an inventory linking named inputs, code, configurations and outputs.

These records support auditability of the saved analyses. They do not establish population prevalence, whole-slide diagnostic performance, external molecular validation or clinical utility.